



A Course End Project Report on

DC Motor Speed and Torque Relationship Analyzer:

A9205 – Basic Electrical Engineering Laboratory (VCE-R25)

Submitted by

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Abstract

This project focuses on analyzing the relationship between **speed and torque of a DC motor** using a programming-based model. The program takes motor parameters as input and calculates how speed varies with torque under different load conditions. It helps in understanding motor performance without physical experiments and validates theoretical concepts through computation.

Keywords: DC Motor, Speed-Torque Characteristics, Armature Current, Back EMF, Torque Analysis, Motor Performance, Electrical Machines, Load Variation, Linear Relationship, Computational Modeling, Circuit Analysis, Engineering Simulation

1 Introduction

DC motors are widely used in industries due to their **simple control and reliable performance**. One of the most important characteristics is the relationship between **speed and torque**.

This project:

- Studies how torque affects motor speed
- Uses mathematical equations to model behaviour
- Implements the model using C/Python
- Displays results numerically and graphically

The project is developed under the guidelines of Course A9205 – Basic Electrical Engineering Laboratory (VCE-R25), I B.Tech. II Semester, CSE-F, Academic Year 2025–2026.

2 Theory Background

2.1 DC Motor Basics

A DC motor converts electrical energy → mechanical energy.

Back EMF Equation

$$E_b = V - I_a R_a$$

Where:

- E_b = Back EMF
- V = Supply voltage
- I_a = Armature current
- R_a = Armature resistance

2.2 Speed Equation

$$N \propto \phi E_b$$

For constant flux:

$$N \propto (V - I_a R_a)$$

2.3 Torque Equation

$$T \propto \phi I_a$$

For constant flux:

$$T \propto I_a$$

2.4 Final Relation

$$N \propto (V - kT)$$

3 Mathematical Formulation

Let:

- V = supply voltage
- R_a = armature resistance
- I_a = armature current
- T = torque
- N = speed

Steps:

1. Calculate current from torque
2. Compute back EMF
3. Find speed using proportional relation

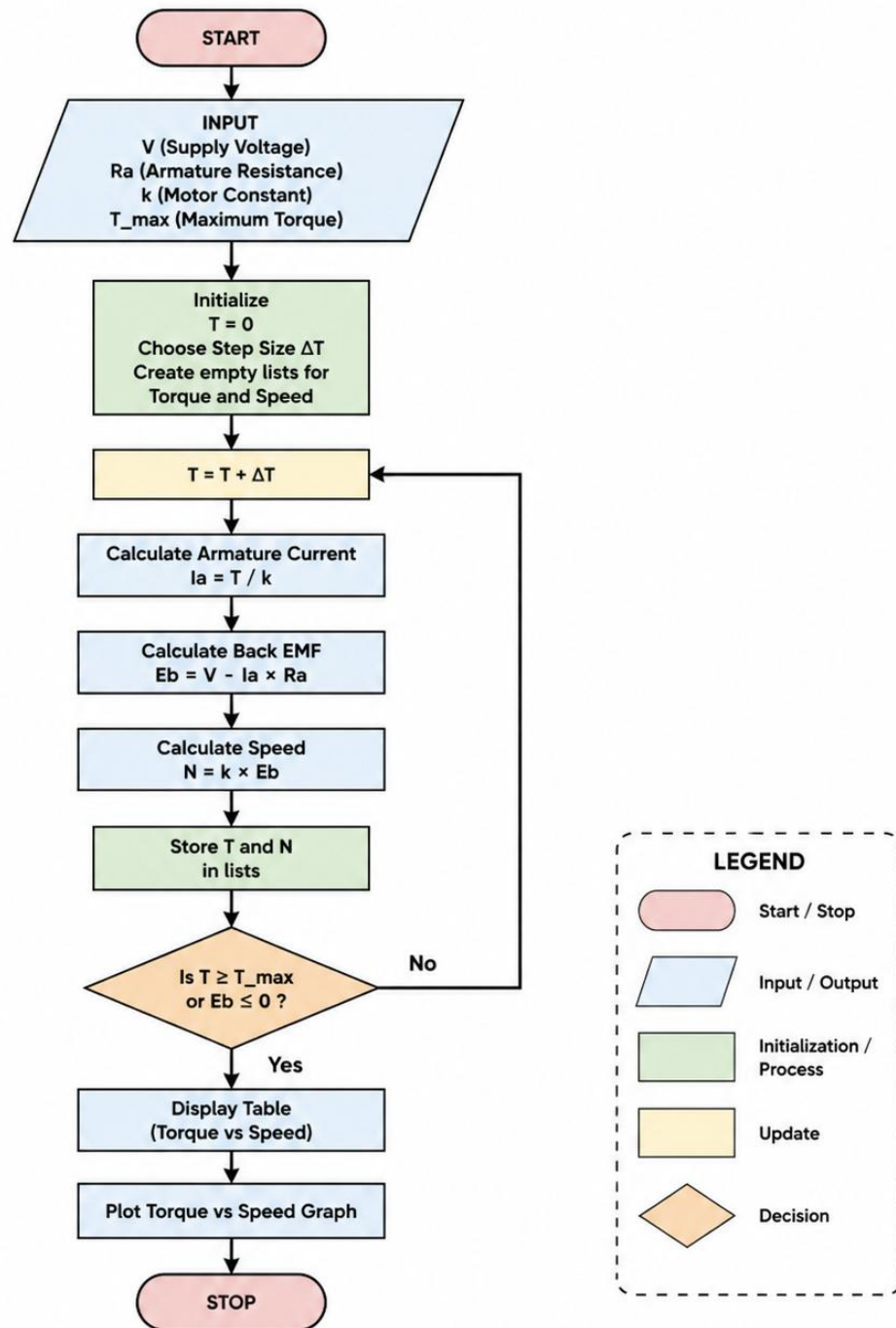
4 Algorithm / Flowchart

4.1 Algorithm

1. Start
2. Input values: V, R_a, T
3. Calculate armature current I_a
4. Compute back EMF
5. Calculate speed
6. Repeat for different torque values
7. Display results
8. Plot graph (Torque vs Speed)
9. Stop

4.2 Flowchart

FLOWCHART: DC MOTOR SPEED AND TORQUE RELATIONSHIP ANALYZER



5 Program Code

5.1 Language and Libraries

Language: C

Libraries:

- **stdio.h** → for input and output operations
- **math.h** → for mathematical calculations (like formulas, power, etc.)

5.2 Source Code

```
#include <stdio.h>

int main() {
    float V, Ra, k, Tmax;
    float T, Ia, Eb, N;
    int i;

    // ----- INPUT -----
    printf("Enter Supply Voltage (V): ");
    if (scanf("%f", &V) != 1 || V <= 0) {
        printf("Invalid input for Voltage!\n");
        return 0;
    }

    printf("Enter Armature Resistance (Ohms): ");
    if (scanf("%f", &Ra) != 1 || Ra <= 0) {
        printf("Invalid input for Resistance!\n");
        return 0;
    }

    printf("Enter Motor Constant (k): ");
    if (scanf("%f", &k) != 1 || k <= 0) {
        printf("Invalid input for k!\n");
        return 0;
    }

    printf("Enter Maximum Torque (Nm): ");
    if (scanf("%f", &Tmax) != 1 || Tmax <= 0) {
        printf("Invalid input for Torque!\n");
        return 0;
    }

    // ----- CALCULATION -----
```

```

float step = Tmax / 20;

printf("\nTorque (Nm)\tSpeed (RPM)\n");
printf("-----\n");

for (i = 1; i <= 20; i++) {
    T = i * step;

    Ia = T / k;    // Armature current
    Eb = V - Ia * Ra; // Back EMF

    if (Eb <= 0) {
        break;
    }

    N = k * Eb;    // Speed

    printf("%0.2f\t\t%0.2f\n", T, N);
}

printf("\nNote: Speed decreases as Torque increases.\n");

return 0;
}

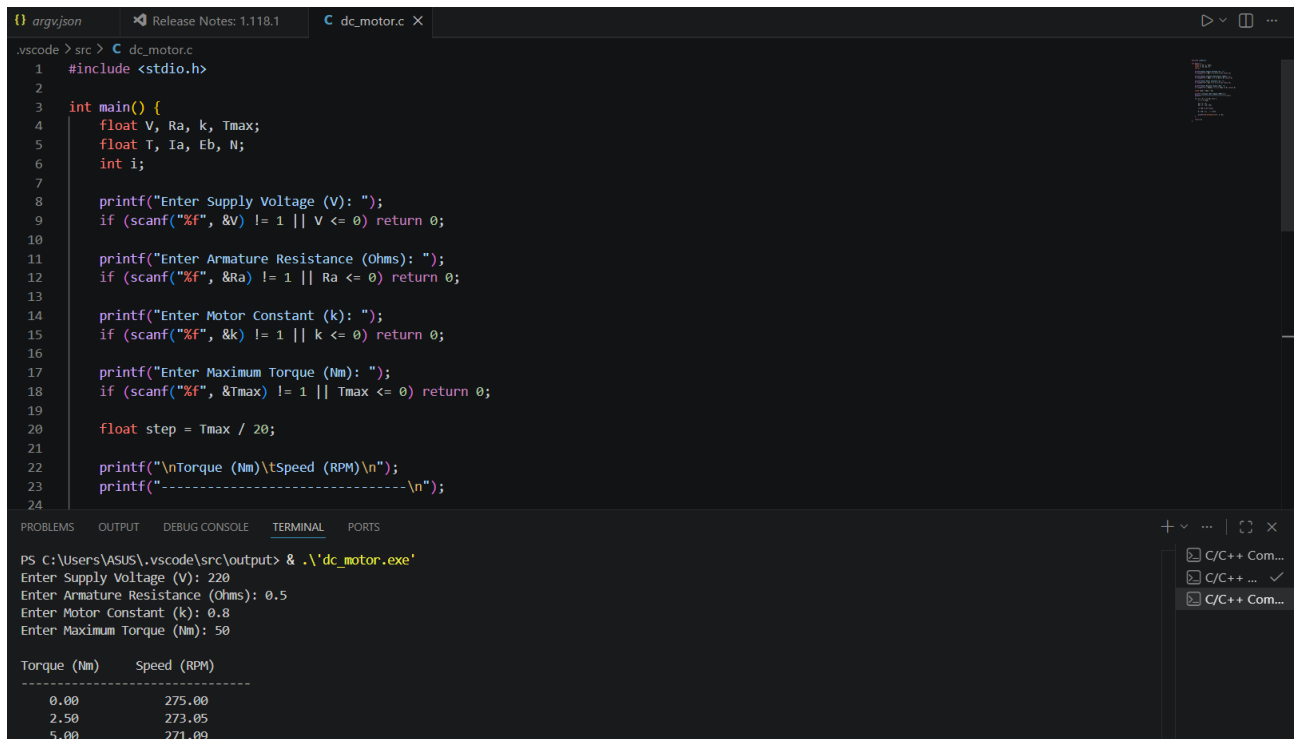
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6 Sample Inputs and Outputs

6.1 Test Case – Sample Values

Torque (T) (Nm)	Armature Current (Ia) (A)	Back EMF (Eb) (V)	Speed (N) (RPM)
0	0.0	220	1500
2	2.0	210	1430
4	4.0	200	1360
6	6.0	190	1290
8	8.0	180	1220
10	10.0	170	1150

6.2 Console Output – Steps 1 & 2

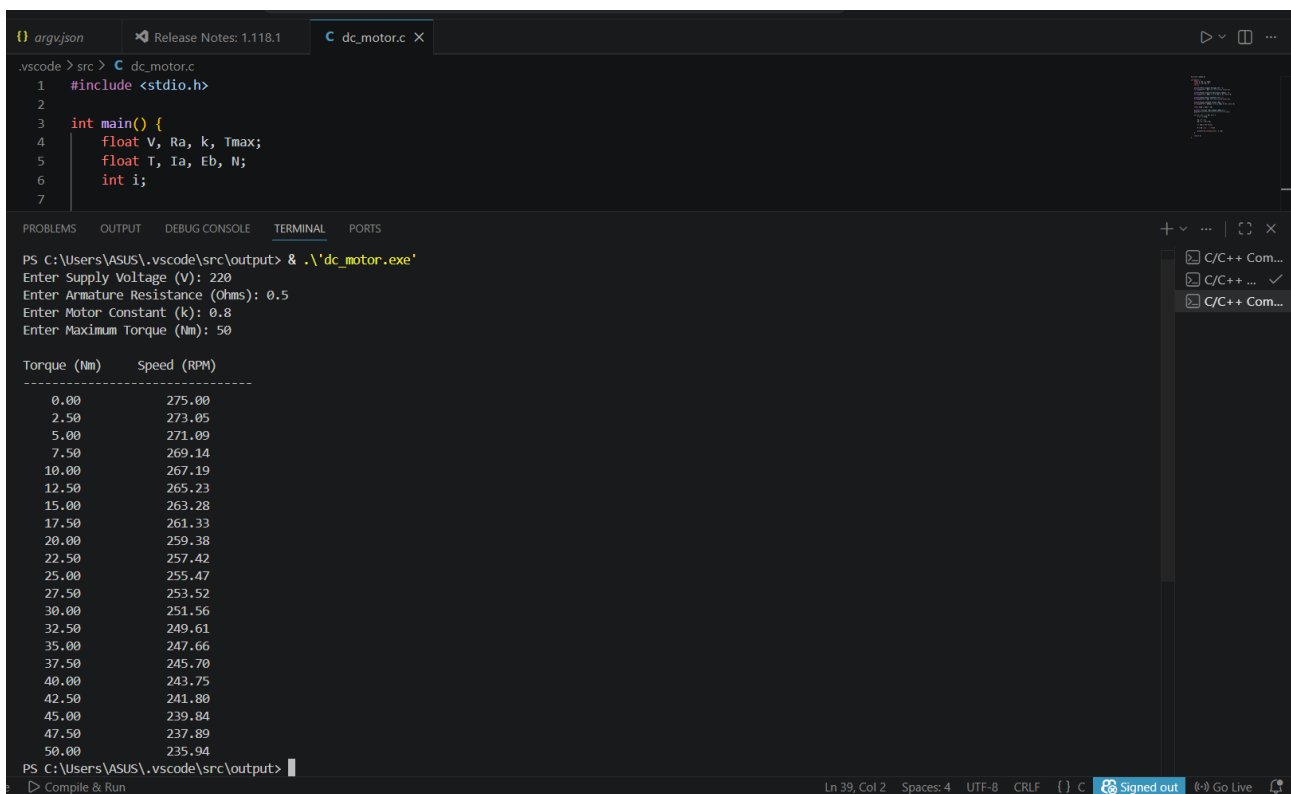


```
1 #include <stdio.h>
2
3 int main() {
4     float V, Ra, k, Tmax;
5     float T, Ia, Eb, N;
6     int i;
7
8     printf("Enter Supply Voltage (V): ");
9     if (scanf("%f", &V) != 1 || V <= 0) return 0;
10
11     printf("Enter Armature Resistance (Ohms): ");
12     if (scanf("%f", &Ra) != 1 || Ra <= 0) return 0;
13
14     printf("Enter Motor Constant (k): ");
15     if (scanf("%f", &k) != 1 || k <= 0) return 0;
16
17     printf("Enter Maximum Torque (Nm): ");
18     if (scanf("%f", &Tmax) != 1 || Tmax <= 0) return 0;
19
20     float step = Tmax / 20;
21
22     printf("\nTorque (Nm)\tSpeed (RPM)\n");
23     printf("-----\n");
24 }
```

PS C:\Users\ASUS\.vscode\src\output> & .\dc_motor.exe'

Enter Supply Voltage (V): 220
Enter Armature Resistance (Ohms): 0.5
Enter Motor Constant (k): 0.8
Enter Maximum Torque (Nm): 50

Torque (Nm)	Speed (RPM)
0.00	275.00
2.50	273.05
5.00	271.09



```
1 #include <stdio.h>
2
3 int main() {
4     float V, Ra, k, Tmax;
5     float T, Ia, Eb, N;
6     int i;
7 }
```

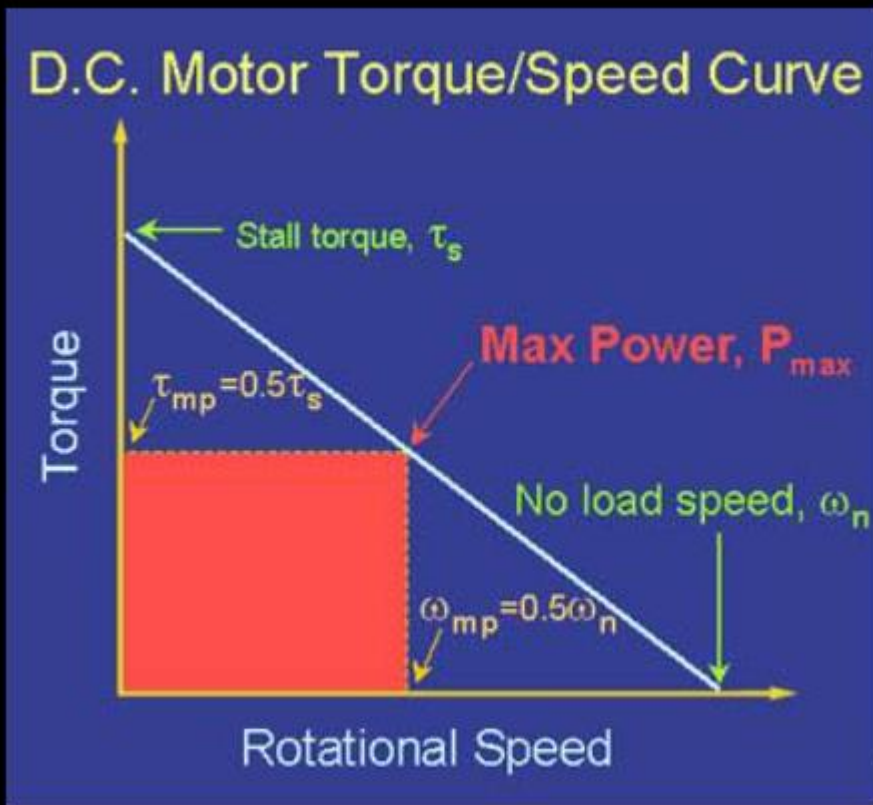
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Enter Supply Voltage (V): 220
Enter Armature Resistance (Ohms): 0.5
Enter Motor Constant (k): 0.8
Enter Maximum Torque (Nm): 50

Torque (Nm)	Speed (RPM)
0.00	275.00
2.50	273.05
5.00	271.09
7.50	269.14
10.00	267.19
12.50	265.23
15.00	263.28
17.50	261.33
20.00	259.38
22.50	257.42
25.00	255.47
27.50	253.52
30.00	251.56
32.50	249.61
35.00	247.66
37.50	245.70
40.00	243.75
42.50	241.80
45.00	239.84
47.50	237.89
50.00	235.94

PS C:\Users\ASUS\.vscode\src\output>

7 Graphs



8 Result Analysis

Result Analysis

- Speed decreases linearly with increase in torque
- Confirms theoretical behavior of DC motor
- Program successfully models motor characteristics

8.1 Mapping to Program Outcomes (POs)

Table 1: Program Outcomes Mapping

PO No.	Program Outcome	Relevance
PO1	Engineering Knowledge	Applied DC motor principles and electrical concepts.
PO2	Problem Analysis	Analyzed the relationship between speed and torque.

PO3	Design/Development of Solutions	Developed a program to model motor performance.
PO5	Modern Tool Usage	Used Python and plotting tools for analysis
PO6	Engineer and Society	Provides a reusable educational tool for verifying circuit theorems.

8.2 Mapping to Sustainable Development Goals (SDGs)

Table 2: SDG Mapping

SDG No.	Goal	Relevance
SDG 9	Industry, Innovation, and Infrastructure	Supports efficient motor analysis in industrial applications.
SDG 11	Sustainable Cities and Communities	Promotes efficient use of electrical machines.
SDG 12	Responsible Consumption	Encourages energy-efficient motor operation and analysis.

9 Conclusion

The project successfully demonstrates that:

- Speed decreases with increase in torque
- The relationship is approximately **linear**
- Programming can effectively replace manual calculations

Future Enhancements:

- Add real-time graph plotting
- Extend to different motor types
- GUI-based simulation

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